The Sharp Logarithmic Coefficient of a Controlled Information Envelope

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Abstract

For finitely many centered, finitely supported increment laws with a common positive variance, we identify the logarithmic displacement of their adaptive lower CDF envelope. Its coefficient is the range of the third moments divided by $3v\sqrt{2\pi e}$. The proof uses the established logarithmic envelope-width bound, an explicit Ornstein–Uhlenbeck corrector, and an integrated stability estimate. It does not differentiate the actual Bellman envelope. The resulting error is $O(\sqrt{\log n \log\log n})$, both in the mean and in unscaled W_1 distance from a translated Gaussian. The finite causal entropy sandwich transfers the coefficient to optimal seed entropy.

Status and dependency. This is an analytic proof manuscript, not a Lean certificate or an independently reviewed paper. It uses the logarithmic moment bound from the accompanying repository as an explicit input, stated below. No priority claim is made. In particular, an $O(1)$ remainder is *not* proved here.

1 Statement

Let $\mu_1, \dots, \mu_d$ be finitely supported probability laws on $\mathbb{R}$, with

$$\mathbb{E}X_i = 0, \qquad \mathbb{E}X_i^2 = v > 0, \qquad \kappa_i = \mathbb{E}X_i^3.$$

Put $\kappa_- = \min_i \kappa_i$, $\kappa_+ = \max_i \kappa_i$ and

$$A_i f(x) = \mathbb{E}f(x - X_i), \quad Tf = \min_i A_i f, \quad Uf = \max_i A_i f, \quad H(x) = \mathbf{1}_{[0,\infty)}(x).$$

The n-step lower and upper policy CDF envelopes are $G_n = T^n H$ and $V_n = U^n H$. Policies may depend on the complete observed history. Their terminal increment sums have CDFs between G_n and V_n. Write ν_n for the law with CDF G_n, and $e_n = \int_{\mathbb{R}}(H - G_n)$ for its mean. Let ϕ, Φ denote the standard normal density and CDF. All logarithms in asymptotic terms are natural.

Theorem 1. *For every fixed family as above, with*

$$c = \frac{\kappa_+ - \kappa_-}{3v\sqrt{2\pi e}}, \tag{1}$$

one has, as $n \to \infty$,

$$W_1\big(\nu_n, N(c\ln n, vn)\big) = O\big(\sqrt{\ln n \ln\ln n}\big), \tag{2}$$

$$e_n = c\ln n + O\big(\sqrt{\ln n \ln\ln n}\big). \tag{3}$$

The constants may depend on the fixed increment family. No lattice, aperiodicity or nonlattice hypothesis is imposed.

Corollary 2 (Causal seed entropy)**.** *For a finite stationary kernel whose action laws P_i have common entropy h and common positive information variance v, set $\kappa_i = \mathbb{E}(-\log_2 P_i(Y_i) - h)^3$. In bits,*

$$\boxed{C_n = nh + \frac{\kappa_+ - \kappa_-}{3v\sqrt{2\pi e}} \ln n + O(\sqrt{\ln n \ln\ln n}).}$$

The same expansion holds for the lower envelope mean and for any exact seeds whose entropy is within a bounded additive amount of that mean.

Proof. Center each action's information by h. The centered envelope is G_n, and $L_n - nh = e_n$. Apply Theorem 1 and the finite-tree sandwich $L_n \le C_n \le L_n + c_\star$ from [1]. □

2 The existing input and integrated stability

Input: logarithmic moment bound. For every finite centered increment family of common positive variance, the repository proves $0 \le e_n \le K(1 + \ln(n+1))$. The declaration is `CausalSpectrum.Sqrt.IncFamily.eG_log_upper` in `MomentAssembly.lean`. Its nondegeneracy assumption is automatic: a finite law of positive variance has two distinct atoms, a positive equal-mass part of which is a translated symmetric two-point law. This input is an order bound, not an assumption of the coefficient to be proved.

Apply the same result to the reflected increment family $(-X_i)_i$. Reflection exchanges the upper and lower envelopes, apart from irrelevant values at their finitely many jump points. Consequently

$$w_n := \int_{\mathbb{R}} (V_n - G_n) = e_n + e_n^{\mathrm{ref}} \le K(1 + \ln(n+1)). \tag{4}$$

In particular $\bar{w}_n := \max_{0\le j\le n} w_j = O(\ln(n+2))$.

Lemma 3 (Integrated amplification)**.** *For $r \in L^1(\mathbb{R}) \cap BV(\mathbb{R})$,*

$$\int_{\mathbb{R}} U^n|r|(x)\,dx \le \|r\|_1 + w_n \mathrm{TV}(r). \tag{5}$$

For bounded functions f, g with $f - g \in L^1 \cap BV$,

$$\|T^n f - T^n g\|_1 \le \|f - g\|_1 + w_n \mathrm{TV}(f - g). \tag{6}$$

Proof. Fix a reference policy repeating one action, and let F_0 be its n-step CDF. For any policy π, its CDF F_π and F_0 lie between G_n and V_n. For absolutely continuous r, integration by parts, with $r(\pm\infty) = 0$, gives

$$\mathbb{E}_\pi|r(x - S_n)| \le \mathbb{E}_0|r(x - S_n)| + \int_{\mathbb{R}} |r'(x-y)|(V_n(y) - G_n(y))\,dy.$$

The bound is independent of π. Take the supremum over policies, integrate in x, and apply Tonelli. This proves (5) for absolutely continuous r. Mollification proves the BV version: variation does not increase; for a fixed horizon there are finitely many policies, so their finite-translation operators are continuous on L^1, and the supremum passes to the limit in L^1. Finally $|T^n f - T^n g| \le U^n|f - g|$ pointwise, which proves (6). □

3 An explicit resonant corrector

Define

$$f(z) = \min_i \left[-\frac{\kappa_i}{6v^{3/2}}(z^2-1)\phi(z) \right], \tag{7}$$

$$a = \int_{\mathbb{R}} f(z)\,dz = -\frac{\kappa_+ - \kappa_-}{3v^{3/2}}\phi(1) = -\frac{c}{\sqrt{v}}, \tag{8}$$

$$g(z) = 2(a\phi(z) - f(z)), \qquad J(z) = \int_{-\infty}^{z} g(y)\,dy, \tag{9}$$

$$B(z) = \phi(z)\int_0^z \frac{J(y)}{\phi(y)}\,dy. \tag{10}$$

Indeed $\int(z^2-1)\phi = 0$ and $\int|z^2-1|\phi = 4\phi(1)$, proving (8). Since $\int g = 0$, formula (10) decays at both ends and solves

$$B'' + zB' + B = 2(a\phi - f). \tag{11}$$

Regularity and tails, explicitly. The function f is continuous, smooth off $\{-1,1\}$, and has only derivative jumps there. Thus $B \in C^2$, with locally absolutely continuous second derivative. For $z < -1$ and $z > 1$, respectively,

$$\frac{J(z)}{\phi(z)} = 2a\frac{\Phi(z)}{\phi(z)} - \frac{\kappa_+ z}{3v^{3/2}}, \qquad \frac{J(z)}{\phi(z)} = -2a\frac{1-\Phi(z)}{\phi(z)} - \frac{\kappa_+ z}{3v^{3/2}}.$$

The elementary Gaussian tail inequality $1-\Phi(z) \le \phi(z)/z$ for $z > 0$, and its reflected version, show that the integral in (10) is $O(1+z^2)$. The equation $B' + zB = J$ and (11), followed by its almost-everywhere derivative, then show that B, B', B'', B''' are bounded by a polynomial times ϕ on each tail. All their polynomially weighted absolute integrals are finite. In particular $B \in W^{3,1}(\mathbb{R})$. No fourth classical derivative of B is required.

For $t \ge 1$, put

$$z = \frac{x - c\ln t}{\sqrt{vt}}, \qquad p_t(x) = \Phi(z), \quad q_t(x) = t^{-1/2}B(z), \quad u_t = p_t + q_t. \tag{12}$$

The function u_t need not be a CDF; it is used only as a bounded comparison profile. Its limiting values are zero and one. Direct differentiation using (11) gives

$$\partial_t u_t - \frac{v}{2}\partial_x^2 u_t = t^{-3/2}f(z) - \frac{c}{\sqrt{v}}\,t^{-2}B'(z). \tag{13}$$

The translation $c\ln t$ absorbs the resonant Gaussian mode exactly.

4 Summable discrete residuals

Lemma 4. *There is a finite K, depending only on the increment family, such that for every real $t \ge 1$, the residual $R_t = u_{t+1} - Tu_t$ belongs to $W^{1,1}(\mathbb{R})$ and*

$$\|R_t\|_1 + \|R_t'\|_1 \le Kt^{-3/2}. \tag{14}$$

Proof. We give the derivative estimates as well as the value estimate, since the derivative is essential to integrated stability. The regularity above and the change of variables $dx = \sqrt{vt}\,dz$ imply

$$\left\|\partial_x^k p_t\right\|_1 = O(t^{-(k-1)/2}) \quad (k \ge 1),$$
$$\left\|\partial_x^k q_t\right\|_1 = O(t^{-k/2}) \quad (0 \le k \le 3),$$
$$\left\|\partial_t^2 u_t\right\|_1 = O(t^{-3/2}), \qquad \|\partial_t\partial_x q_t\|_1 = O(t^{-3/2}).$$

For the last two estimates, use $\partial_t z = -z/(2t) - c/(\sqrt{v}\, t^{3/2})$; only weighted B, B', B'' occur. Also $\partial_t p_t = (v/2)p_t'' - (c/t)p_t'$. All O constants here are finite and uniform for $t \geq 1$.

Taylor's formula under translation, in L^1, gives for each action

$$A_i p_t = p_t + \frac{v}{2}p_t'' - \frac{\kappa_i}{6}p_t''' + e_{i,t}, \qquad \|e_{i,t}\|_1 \leq \frac{\mathbb{E}|X_i|^4}{24}\,\|p_t''''\|_1 = O(t^{-3/2}),$$

and

$$A_i q_t = q_t + \frac{v}{2}q_t'' + b_{i,t}, \qquad \|b_{i,t}\|_1 \leq \frac{\mathbb{E}|X_i|^3}{6}\,\|q_t'''\|_1 = O(t^{-3/2}).$$

The scalar minimum is Lipschitz with respect to the maximum norm. Since $\min_i[-\kappa_i p_t'''/6] = t^{-3/2}f(z)$, summing these error bounds yields

$$\left\|Tu_t - u_t - \tfrac{v}{2}u_t'' - t^{-3/2}f(z)\right\|_1 = O(t^{-3/2}).$$

Taylor expansion in time, (13), and $\|t^{-2}B'(z)\|_1 = O(t^{-3/2})$ prove the estimate for R_t.

For its derivative, write $d_{i,t} = A_i u_t - u_t - (v/2)u_t''$. Taylor expansion to order two for p_t' gives

$$\|A_i p_t' - p_t' - \tfrac{v}{2}p_t'''\|_1 \leq \frac{\mathbb{E}|X_i|^3}{6}\,\|p_t''''\|_1 = O(t^{-3/2}).$$

Centered Taylor expansion to order one for q_t', followed by the triangle inequality, gives

$$\|A_i q_t' - q_t' - \tfrac{v}{2}q_t'''\|_1 \leq v\,\|q_t'''\|_1 = O(t^{-3/2}).$$

Hence $\left\|d_{i,t}'\right\|_1 = O(t^{-3/2})$. The minimum of finitely many $W^{1,1}$ functions has derivative bounded almost everywhere by the sum of the absolute derivatives. Finally

$$\|p_{t+1}' - p_t' - \tfrac{v}{2}p_t'''\|_1 = O(t^{-3/2}).$$

To verify this last estimate, integrate $\partial_s p_s' = (v/2)p_s''' - (c/s)p_s''$ on $[t, t+1]$; $\|p_s''\|_1 = O(s^{-1/2})$ and $\|\partial_s p_s'''\|_1 = O(s^{-2})$. For q_t, the preceding time-derivative bound and $\|q_t'''\|_1 = O(t^{-3/2})$ give the same estimate. Differentiating $R_t = u_{t+1} - u_t - (v/2)u_t'' - \min_i d_{i,t}$ now proves the derivative bound. All derivatives in this paragraph may be read as weak derivatives, so the two switching points cause no missing term. □

5 From a late-start profile to the actual envelope

For any $t_0 \geq 1$, successive one-step comparisons and positivity of U give the pointwise bound

$$|T^n u_{t_0} - u_{t_0+n}| \leq \sum_{j=0}^{n-1} U^{n-1-j}|R_{t_0+j}|.$$

Here one uses $|Tf - Tg| \leq U|f - g|$ and subadditivity of U. By Lemmas 3 and 4,

$$\|T^n u_{t_0} - u_{t_0+n}\|_1 \leq K\frac{1+\bar{w}_n}{\sqrt{t_0}}. \tag{15}$$

Thus a summable residual is sufficient after its derivative's amplification by the envelope width has also been accounted for.

It remains to compare the step initial condition with a Gaussian. Let $s = \sqrt{vt_0}$, $b(x) = \Phi(x/s)$, choose $R > 0$, and put $A = sR$, $\varepsilon = 1 - \Phi(R)$. The pointwise inequalities

$$H(x) \leq b(x + A) + r_+(x), \qquad b(x - A) \leq H(x) + r_-(x)$$

hold with $r_+(x) = \mathbf{1}_{x\geq 0}(1 - b(x + A))$ and $r_-(x) = \mathbf{1}_{x<0}b(x - A)$. They satisfy

$$\|r_\pm\|_1 \leq s\phi(R), \qquad \mathrm{TV}(r_\pm) = 2\varepsilon.$$

Propagate the inequalities with T, use translation invariance and Lemma 3, and integrate the positive and negative parts. Since $T^n b$ is a CDF, its integral difference from a translate by A is exactly A. Therefore

$$\|G_n - T^n b\|_1 \leq 2A + 2s\phi(R) + 4\varepsilon w_n. \tag{16}$$

The Gaussian p_{t_0} is b translated by $c\ln t_0$, so $\|T^n p_{t_0} - T^n b\|_1 = |c\ln t_0|$. Furthermore $\|q_{t_0}\|_1 = \sqrt{v}\,\|B\|_1$ and $\mathrm{TV}(q_{t_0}) = t_0^{-1/2}\,\|B'\|_1$. Thus

$$\|T^n u_{t_0} - T^n p_{t_0}\|_1 \leq \sqrt{v}\,\|B\|_1 + w_n t_0^{-1/2}\,\|B'\|_1\,. \tag{17}$$

Take $t_0 = \ln(n + e^2)$ and $R = \sqrt{2\ln t_0}$. Then $\phi(R) = (2\pi)^{-1/2}/t_0$, $\varepsilon \leq \phi(R)/R$, and $\bar{w}_n = O(t_0)$ by (4). Equations (15)–(17) imply

$$\|G_n - u_{n+t_0}\|_1 = O(\sqrt{\ln n \ln\ln n}). \tag{18}$$

Since $\|u_t - p_t\|_1 = \sqrt{v}\,\|B\|_1$, this also holds with p_{n+t_0} in place of u_{n+t_0}. Coupling Gaussians with the same standard normal variable shows

$$W_1\big(N(c\ln(n+t_0), v(n+t_0)), N(c\ln n, vn)\big) \leq |c|\ln(1+t_0/n) + \sqrt{2v/\pi}\,(\sqrt{n+t_0} - \sqrt{n}) = o(1).$$

The one-dimensional identity $W_1(\mu,\nu) = \int |F_\mu - F_\nu|$ proves (2). The difference of means is at most W_1, proving (3) and completing Theorem 1. □

6 A concrete dyadic information pair

Take

$$P = (1/4, \underbrace{1/16, \ldots, 1/16}_{12}), \qquad Q = (\underbrace{1/8, \ldots, 1/8}_{6}, \underbrace{1/32, \ldots, 1/32}_{8}).$$

Their entropies are $h = 7/2$, their information variances are $v = 3/4$, and their centered third moments are $-3/4$ and $3/4$. Every atom is a power of $1/2$, so dyadic exactness gives $C_n = L_n$ at every horizon. Hence

$$\boxed{C_n = \frac{7}{2}n + \frac{2}{3\sqrt{2\pi e}}\ln n + O(\sqrt{\ln n \ln\ln n}).}$$

The coefficient is $0.161313816346\ldots$ bits per unit of $\ln n$. The result is about the unrestricted optimal causal seed, not a particular block construction.

Scope. Only the extreme third moments enter the logarithmic coefficient. Intermediate actions and higher moments can affect the remainder. When the third moments all agree, the existing moment classification already gives the stronger bounded-excess conclusion; this manuscript is principally useful when $\kappa_+ > \kappa_-$. It does not prove a bounded remainder, a pointwise $1/n$ limit for $e_{n+1} - e_n$, or a uniform theorem as $v \downarrow 0$ or the kernel varies. The profiles u_t are auxiliary functions, not asserted probability laws.

Related work and provenance. Rates in martingale central limit theorems and nonlinear central limit theorems have a substantial literature; see [2, 3]. Those references provide context, not the coefficient calculation in this proof. The nonstandard analytic input here is specifically the *integrated envelope-width* bound (4), supplied by the repository's moment theorem. A bound for each fixed policy in W_1 alone would not justify taking the pointwise envelope.

References

[1] J. Mikulík, *Finite Shannon theorem*, https://github.com/jendamikulik/finite-Shannon-theorem, retrieved 12 October 2026. Analytic input: `CausalSeed/MomentAssembly.lean`, theorem `Sqrt.IncFamily.eG_log_upper`; compare also `MomentWidth.lean`, theorem `polE_abs_le`. The new corrector and coefficient theorem in this manuscript are not claimed to be formalized there.

[2] J.-C. Mourrat, On the rate of convergence in the martingale central limit theorem, *Bernoulli* **19**(2) (2013), 633–645. doi:10.3150/12-BEJ417.

[3] X. Fang, S. Peng, Q.-M. Shao and Y. Song, *Limit theorems with rate of convergence under sublinear expectations*, https://arxiv.org/abs/1711.10649.